Sanjid Hasan

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Research Profile

Electrical and Electronic Engineering undergraduate working on efficient Bengali speech recognition, tokenizer and vocabulary adaptation, model inference, and resource-constrained deployment, open-source speech datasets, reproducible accuracy-latency benchmarking, AI research, and student research leadership.

Education

Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh Expected Oct. 2027
B.Sc. in Electrical and Electronic Engineering

Research & Model-Efficiency Work

NeurIPS Speech Accessibility Challenge – Track 2 Ongoing
Streaming ASR under CPU-only inference constraints

- Lead a six-member university team investigating practical inference, streaming decoding, and deployment decisions for accessible, resource-constrained speech recognition.
- Experiment with sherpa-onnx, ONNX quantization, and alternative decoding strategies, focusing on accuracy, latency, and runtime constraints.

Tokenizer Transplantation: Mitigating Autoregressive Collapse in Edge-Efficient Bengali ASR
Sanjid Hasan, Md. Abdur Rahman — MusIML Workshop, ICML 2026 (Poster); arXiv:2607.09598 2026

- Adapted a 61.5M-parameter Moonshine ASR model to Bengali by transplanting the decoder vocabulary to BanglaBERT WordPiece and resizing embedding/output layers.
- Reduced tokenizer fertility from 9.16 to 1.30 tokens/word and autoregressive sequence length by 85.8%, eliminating the observed cross-script decoding collapse; achieved 21.54% WER at 0.0053 RTF on the Lipi-Ghor test set using a Kaggle T4 GPU.

Make It Hard to Hear, Easy to Learn: Long-Form Bengali ASR and Speaker Diarization via Extreme Augmentation and Perfect Alignment
Sanjid Hasan, Risalat Labib, A H M Fuad, Bayazid Hasan — arXiv:2602.23070 2026

- Co-developed a two-stage Whisper-Medium pipeline: full Bengali fine-tuning, followed by 8-bit loading and parameter-efficient fine-tuning with only 1.22% of 773M parameters trainable.
- Converted the model to CTranslate2 FP16 for faster-whisper inference, cutting processing time for a 22-hour test set from approximately 4 hours to 26 minutes (roughly 10× faster; approximately 0.019 RTF on Kaggle T4 GPUs).
- Excluded Demucs preprocessing when its small WER gain increased RTF from 0.019 to 0.068. Associated work earned 2nd Runner-Up and the Dataset Prize at BUET CSE Fest DL Sprint 4.0.

Lipi-Ghor-bn-882-SSTT – Open Bengali Speech Resource 2026
doi:10.57967/hf/7877

- Co-created and released an approximately 882-hour, multi-domain Bengali corpus from 1,019 videos across 596 channels, with transcription, diarization, and timestamp annotations for speech research.
- Associated data paper accepted to the SALMA Workshop at EMNLP 2026.

Additional Research & Open-Source Experience

Research Assistant – Hackules Institute for Advanced Research Jan. 2025 – Aug. 2025
Computer vision, generative AI, and AI-based skeletal diagnostic research

- Worked on AI and machine-learning research across computer vision, generative methods, and skeletal diagnostics.
- Fine-tuned transformer models for Bengali speech recognition, emotion and sentiment classification, and other NLP tasks; published research assets through Hugging Face, GitHub, and Kaggle.

Leadership & Campus Experience

Founder – Team Frozen Voltage 2025 – Present
Khulna University of Engineering & Technology

- Founded a student team for electronics, AI, robotics, and technical competitions; organized a free 3–4 week PCB Design & Etching Workshop for KUET Batch 2K24.
- Supported projects including ESP32–AD8232 ECG monitoring with a Streamlit dashboard, RC-plane prototyping, 3D printing, and robotics.

Assistant Domain Lead of NLP & Founding Member – KMinds Oct. 2025 – Present
Student-led AI research community at KUET

- Mentor student work on NLP, speech projects, research preparation, and competitions; organize workshops, and research discussions.
- Lead a six-member team for the NeurIPS Speech Accessibility Challenge.

Technical Secretary – IEEE Student Branch KUET Sept. 2026 – Present
Khulna University of Engineering & Technology

- Manage official communications and applications, coordinate with the executive team, and support seminars and industrial visits.

Technical Secretary & Event Organizer – Techforge KUET Ongoing
Formerly EEE Makers Hub

- Helped organize Circuit Master 1.0, MATLAB workshops, drone sessions, hardware workshops, competitions, and student events.

Assistant Class Representative – Department of EEE, KUET Mar. 2024 – Dec. 2024
Khulna University of Engineering & Technology

- Coordinated class communication and departmental activities for more than 100 students.

Selected Campus Ambassador Experience 2024 – 2026

- **Machine Mania 3.0 – BUTEX TexMech Society:** Represented KUET and supported campus outreach for the national technology festival.
- **SciBlitz 1.0:** Campus Ambassador; received Best Campus Ambassador recognition.

Selected Competition Results & Recognition

- **BUET CSE Fest DL Sprint 4.0 (2026):** 2nd Runner-Up and Dataset Prize, Team Villagers.
- **KUET University Day Technical Poster (Sept. 2026):** 2nd Prize for the Tokenizer Transplantation work.
- **AgriYield (2025):** 3rd Place, solo crop-yield prediction submission.
- **NWU CSE Fest Datathon (2025):** 4th Place, Team Maverics.
- **First Byte Datathon (2025):** 8th Place (Top 10), Team Frozen Voltage.
- **Research acceptances:** MusIML Workshop at ICML 2026 (Poster); SALMA Workshop at EMNLP 2026.

Technical Skills

Programming & ML: Python, PyTorch, Scikit-learn, AutoGluon, Hugging Face Transformers, Jupyter

Efficient Training & Inference: PEFT, bitsandbytes 8-bit loading, FP16, ONNX, sherpa-onnx, CTranslate2, faster-whisper, RTF/latency benchmarking

Speech & NLP: ASR, speaker diarization, tokenizer/vocabulary adaptation, transformer fine-tuning, WER/CER evaluation, dataset curation and alignment

Tools & Hardware: Git, GitHub, Hugging Face, Kaggle, Google Colab, Streamlit, PCB design fundamentals, ESP32 prototyping

Certifications & Additional Recognition

Kaggle Datasets Expert · Best Campus Ambassador – SciBlitz 1.0 · Google Certificates: Transformer Models and BERT; Encoder–Decoder Architecture · CS50's Introduction to Programming with Python (Harvard University)